Table B2. Spline-indicator fixed-parameter binary logit model of crash injury severity (serious injury/fatal vs. PDO/minor injury), crashes involving a motorcycle and another vehicle type, Thailand, 2016–2023: implied parameter for each year, estimated parameter (*t*-statistic).

Variable description	Spline terms retained (*j*)	2016	2017	2018	2019	2020	2021	2022	2023
Constant [serious injury / fatal]									
Constant	1, 3, 4, 8	0.254 (3.82)	0.507 (3.82)	0.132 (0.90)	0.022 (0.16)	–0.087 (–0.59)	–0.197 (–1.14)	–0.306 (–1.46)	0.212 (0.80)
Vehicle involvement									
One motorcycle involved (1 if yes, 0 otherwise)	1, 6, 7	–0.124 (–4.86)	–0.249 (–4.86)	–0.373 (–4.86)	–0.498 (–4.86)	–0.622 (–4.86)	–0.115 (–0.85)	–0.072 (–0.49)	–0.029 (–0.14)
One passenger car involved (1 if yes, 0 otherwise)	1, 2, 6	–0.518 (–5.63)	–0.444 (–6.19)	–0.371 (–5.85)	–0.297 (–4.17)	–0.224 (–2.45)	–0.398 (–5.39)	–0.573 (–6.41)	–0.748 (–5.93)
One pickup truck involved (1 if yes, 0 otherwise)	1, 3, 4, 5, 6	–0.167 (–2.90)	–0.333 (–2.90)	0.064 (0.61)	–0.063 (–0.64)	0.234 (2.42)	0.097 (1.26)	–0.040 (–0.44)	–0.177 (–1.40)
Two or more pickup trucks involved (1 if yes, 0 otherwise)	1, 3, 4, 5, 7, 8	–0.422 (–2.39)	–0.843 (–2.39)	0.232 (0.71)	–0.783 (–2.50)	–0.492 (–2.66)	–0.200 (–0.74)	–0.986 (–2.91)	–0.437 (–1.57)
One 6-wheel truck involved (1 if yes, 0 otherwise)	1, 4, 5, 6	0.443 (5.69)	0.887 (5.69)	1.330 (5.69)	0.670 (2.56)	1.056 (5.76)	0.849 (6.67)	0.642 (4.69)	0.435 (2.15)
One 10-wheel truck involved (1 if yes, 0 otherwise)	1, 2, 6	0.929 (4.26)	1.030 (6.46)	1.132 (8.91)	1.233 (8.82)	1.335 (7.07)	1.142 (8.35)	0.949 (5.85)	0.756 (3.12)
One truck with more than 10 wheels involved (1 if yes, 0 otherwise)	1, 2	1.229 (8.51)	1.226 (10.42)	1.223 (12.78)	1.220 (14.78)	1.217 (14.76)	1.214 (12.71)	1.211 (10.31)	1.208 (8.38)
Van involved (1 if yes, 0 otherwise)	1, 8	0.052 (1.98)	0.104 (1.98)	0.157 (1.98)	0.209 (1.98)	0.261 (1.98)	0.313 (1.98)	0.365 (1.98)	–0.670 (–1.86)
Crash type									
Rear-end collision in the same lane (1 if yes, 0 otherwise)	1, 2, 5, 7	–0.357 (–4.08)	–0.249 (–4.12)	–0.141 (–2.36)	–0.034 (–0.39)	–0.133 (–2.16)	–0.232 (–2.48)	–0.111 (–1.50)	0.009 (0.08)
Sideswipe (parallel movement) (1 if yes, 0 otherwise)	1, 2, 3	–0.720 (–5.53)	–0.422 (–4.26)	–0.470 (–6.06)	–0.517 (–8.46)	–0.565 (–10.24)	–0.613 (–9.81)	–0.661 (–8.31)	–0.709 (–6.98)
Head-on collision (1 if yes, 0 otherwise)	1, 2	0.870 (4.87)	0.879 (6.11)	0.889 (7.85)	0.898 (9.90)	0.907 (10.87)	0.916 (9.65)	0.926 (7.72)	0.935 (6.16)
Crash cause									
Speeding (1 if yes, 0 otherwise)	1, 4, 5, 6, 8	–0.120 (–4.07)	–0.239 (–4.07)	–0.359 (–4.07)	0.078 (0.77)	–0.073 (–0.70)	0.028 (0.28)	0.128 (0.99)	–0.271 (–1.88)
Cutting in too closely (1 if yes, 0 otherwise)	1, 8	0.012 (0.69)	0.023 (0.69)	0.035 (0.69)	0.046 (0.69)	0.058 (0.69)	0.069 (0.69)	0.081 (0.69)	–0.224 (–1.48)
Running a red light (1 if yes, 0 otherwise)	1, 2, 3, 4	1.005 (2.77)	–0.047 (–0.16)	0.970 (4.49)	0.807 (4.91)	0.644 (4.89)	0.480 (3.63)	0.317 (1.91)	0.154 (0.71)
Roadway characteristics									
Crash on the main carriageway (1 if yes, 0 otherwise)	1, 5, 7	–0.062 (–1.58)	–0.124 (–1.58)	–0.186 (–1.58)	–0.248 (–1.58)	0.159 (1.34)	0.566 (2.84)	0.014 (0.10)	–0.538 (–2.51)
Crash on a frontage road (1 if yes, 0 otherwise)	1, 2, 7, 8	–1.659 (–8.51)	–1.378 (–9.43)	–1.097 (–9.90)	–0.817 (–7.85)	–0.536 (–4.11)	–0.255 (–1.45)	–1.327 (–5.84)	–0.799 (–4.19)
Road under construction (1 if yes, 0 otherwise)	1, 8	0.009 (0.37)	0.017 (0.37)	0.026 (0.37)	0.035 (0.37)	0.043 (0.37)	0.052 (0.37)	0.061 (0.37)	–0.809 (–3.10)
Painted median (1 if yes, 0 otherwise)	1, 5	0.036 (1.57)	0.072 (1.57)	0.108 (1.57)	0.144 (1.57)	0.026 (0.39)	–0.091 (–1.36)	–0.209 (–2.31)	–0.327 (–2.61)
Raised or earth median (1 if yes, 0 otherwise)	1, 2, 6, 7	–0.550 (–6.03)	–0.422 (–6.23)	–0.293 (–5.27)	–0.164 (–2.65)	–0.035 (–0.43)	–0.338 (–3.61)	–0.265 (–3.73)	–0.192 (–1.79)
Depressed median (1 if yes, 0 otherwise)	1, 3, 4	–0.009 (–0.16)	–0.019 (–0.16)	0.329 (3.45)	0.315 (4.39)	0.301 (5.23)	0.288 (4.79)	0.274 (3.53)	0.260 (2.53)
Barrier median (1 if yes, 0 otherwise)	1, 2, 4	–1.233 (–7.70)	–0.969 (–9.80)	–0.705 (–5.68)	–0.735 (–7.79)	–0.765 (–10.00)	–0.795 (–10.08)	–0.825 (–8.25)	–0.855 (–6.52)
Horizontal curve (1 if yes, 0 otherwise)	1, 2	0.348 (3.00)	0.315 (3.37)	0.283 (3.76)	0.250 (3.95)	0.218 (3.49)	0.185 (2.55)	0.152 (1.69)	0.120 (1.07)
Concrete pavement surface (1 if yes, 0 otherwise)	1, 4, 5, 7, 8	–0.194 (–5.08)	–0.388 (–5.08)	–0.581 (–5.08)	–0.106 (–0.88)	–0.373 (–5.14)	–0.641 (–5.79)	0.085 (0.69)	–0.261 (–1.97)
T-intersection (1 if yes, 0 otherwise)	1, 3, 4, 5	–0.119 (–1.48)	–0.238 (–1.48)	0.377 (2.14)	–0.231 (–1.64)	–0.142 (–1.41)	–0.052 (–0.67)	0.037 (0.41)	0.126 (1.00)
Four-leg intersection (1 if yes, 0 otherwise)	1, 3, 4, 6	–0.002 (–0.02)	–0.003 (–0.02)	0.588 (3.18)	0.212 (1.95)	–0.164 (–1.13)	–0.139 (–1.40)	–0.114 (–1.10)	–0.089 (–0.58)
U-turn without auxiliary lane (1 if yes, 0 otherwise)	1, 2	0.403 (2.21)	0.354 (2.41)	0.304 (2.59)	0.254 (2.51)	0.205 (1.95)	0.155 (1.23)	0.105 (0.66)	0.056 (0.28)
U-turn with deceleration lane only (1 if yes, 0 otherwise)	1, 2	0.682 (3.51)	0.596 (3.75)	0.510 (3.99)	0.424 (4.04)	0.338 (3.51)	0.252 (2.38)	0.166 (1.28)	0.079 (0.49)
U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise)	1, 2, 3	0.828 (2.50)	0.114 (0.56)	0.186 (1.19)	0.259 (2.06)	0.331 (2.72)	0.404 (2.75)	0.476 (2.51)	0.549 (2.27)
Public or commercial access point (1 if yes, 0 otherwise)	1	0.011 (0.68)	0.022 (0.68)	0.033 (0.68)	0.044 (0.68)	0.054 (0.68)	0.065 (0.68)	0.076 (0.68)	0.087 (0.68)
Environmental characteristics									
Wet road surface (1 if yes, 0 otherwise)	1, 2, 3, 4	–0.164 (–0.26)	18.322 (35.90)	0.216 (0.45)	0.076 (0.21)	–0.063 (–0.25)	–0.203 (–1.00)	–0.343 (–1.44)	–0.483 (–1.44)
Raining (1 if yes, 0 otherwise)	1, 2, 3, 4	1.063 (1.56)	–17.666 (–34.59)	0.059 (0.12)	0.125 (0.33)	0.192 (0.71)	0.258 (1.18)	0.325 (1.24)	0.392 (1.07)
Dark, no street lighting (1 if yes, 0 otherwise)	1, 2, 7	0.739 (5.75)	0.767 (7.35)	0.795 (8.08)	0.823 (7.24)	0.851 (5.93)	0.878 (4.86)	0.573 (3.28)	0.268 (1.17)
Dark, with street lighting (1 if yes, 0 otherwise)	1, 4, 5, 6	0.108 (3.14)	0.216 (3.14)	0.325 (3.14)	–0.069 (–0.54)	0.220 (1.67)	0.198 (1.45)	0.176 (1.12)	0.154 (0.82)
Temporal characteristics									
Weekend (1 if yes, 0 otherwise)	1	0.010 (1.37)	0.019 (1.37)	0.029 (1.37)	0.038 (1.37)	0.048 (1.37)	0.057 (1.37)	0.067 (1.37)	0.076 (1.37)
Songkran festival period (1 if yes, 0 otherwise)	1, 2, 3	0.097 (0.76)	–0.425 (–4.62)	–0.405 (–5.57)	–0.386 (–6.50)	–0.366 (–6.56)	–0.347 (–5.42)	–0.327 (–4.08)	–0.308 (–3.05)
New Year festival period (1 if yes, 0 otherwise)	1, 5	–0.127 (–6.49)	–0.254 (–6.49)	–0.381 (–6.49)	–0.509 (–6.49)	–0.501 (–8.65)	–0.493 (–8.63)	–0.485 (–6.33)	–0.478 (–4.51)
Public holiday (1 if yes, 0 otherwise)	1, 7	–0.037 (–1.11)	–0.073 (–1.11)	–0.110 (–1.11)	–0.147 (–1.11)	–0.184 (–1.11)	–0.220 (–1.11)	0.196 (1.22)	0.612 (2.17)
Evening (1 if yes, 0 otherwise)	1	0.029 (1.28)	0.058 (1.28)	0.087 (1.28)	0.116 (1.28)	0.145 (1.28)	0.174 (1.28)	0.204 (1.28)	0.233 (1.28)

Variable description	Spline terms retained (*j*)	2016	2017	2018	2019	2020	2021	2022	2023
Midnight / early morning (1 if yes, 0 otherwise)	1, 5, 8	0.214 (6.87)	0.428 (6.87)	0.642 (6.87)	0.856 (6.87)	0.705 (6.14)	0.554 (3.97)	0.402 (2.17)	0.843 (3.83)
Peak hours (1 if yes, 0 otherwise)	1	0.003 (0.42)	0.007 (0.42)	0.010 (0.42)	0.013 (0.42)	0.016 (0.42)	0.020 (0.42)	0.023 (0.42)	0.026 (0.42)
Region									
Eastern region (1 if yes, 0 otherwise)	1, 2, 6, 8	1.012 (6.45)	1.032 (9.18)	1.051 (12.02)	1.070 (10.87)	1.089 (7.95)	0.823 (8.72)	0.557 (3.76)	0.901 (5.14)
Northeastern region (1 if yes, 0 otherwise)	1, 2, 3, 6, 7	0.779 (6.71)	0.630 (6.25)	0.815 (11.95)	1.000 (15.03)	1.184 (12.17)	0.858 (7.87)	0.936 (12.48)	1.014 (8.53)
Northern region (1 if yes, 0 otherwise)	1, 5, 6, 7	0.114 (4.10)	0.229 (4.10)	0.343 (4.10)	0.458 (4.10)	–0.287 (–2.53)	–0.506 (–4.26)	–0.279 (–3.28)	–0.053 (–0.38)
Southern region (1 if yes, 0 otherwise)	1, 2, 5	–0.049 (–0.46)	0.185 (2.54)	0.418 (6.25)	0.652 (6.98)	0.600 (8.85)	0.547 (8.95)	0.495 (6.29)	0.442 (4.05)
Western region (1 if yes, 0 otherwise)	1, 2, 6	0.450 (3.21)	0.548 (5.40)	0.646 (8.03)	0.744 (8.25)	0.843 (6.81)	0.732 (8.41)	0.622 (6.09)	0.512 (3.31)
Model statistics									
Number of observations					21,371				
Number of estimated spline coefficients					143				
Variables with the base term only (parameter proportional to the period index)					4 of 46				
Log-likelihood at zero, LL(0)					–14813.25				
Log-likelihood with constant only, LL(C)					–14745.16				
Log-likelihood at convergence, LL(β)					–12123.32				
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$					0.182				
AIC / BIC					24,532.6 / 25,672.3				
LR test against the fixed-parameter separate-period models					$\chi^2 = 142.35$, df = 225, confidence of rejection = 0.000				

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median. Following Shabab et al. (2024), each variable enters through spline terms $f_j(\tau) = \max(\tau - j + 1, 0)$, $j = 1, \ldots, 8$, where $\tau = 1, \ldots, 8$ indexes the years; the parameter in year τ is $\beta(\tau) = \sum_j b_j f_j(\tau)$. The base term ($j = 1$) is always retained; the other terms were removed by backward elimination at the 90% level ($|t| < 1.645$, one term per variable per round). A variable with the base term only has a parameter proportional to the period index. The table reports the implied parameters $\beta(\tau)$; *t*-statistics by the delta method from the covariance of the spline coefficients.